

Tide retrospective: *generic- (k_1, k_2) 2D mixed-even moment asymptotic*

Daniel Murfet (with Claude Opus 4.7)

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Abstract

Mirror-shape follow-up to the kthkth-partition-asymptotic tide (also 21 May), explicitly named in that tide's Follow-ups. Lifts the V2 (2D moment) machinery from the $(k_1, k_2) = (2, 3)$ quartic-sextic case to parametric (k_1, k_2) with $k_1, k_2 \geq 1$. The four-step packaging (factorisation $\rightarrow$ const-mul-rpow $\rightarrow$ rescaled tendsto $\rightarrow$ **IsEquivalent**) mirrors the partition-asymptotic tide line-for-line; the only material difference is the 2D factorisation step, which now uses **gibbsExpectation_separable_addSeparable** to factor the mixed-even moment into a product of two 1D moments. Build clean first-try; 155 lines.

1 Setting

The kthkth-partition tide (**tide/kthkth-partition-asymptotic**, landed earlier in the same day) lifted the Z2 partition asymptotic from the quartic-sextic specialisation to parametric (k_1, k_2) . Its retrospective named two strict-improvement follow-ups: the generic numerator asymptotic and the generic moment asymptotic. This tide closes the latter. The numerator analogue is left for a future mechanical-extension session.

2 The thing formalised

For $k_1, k_2 \geq 1$, $j_1, j_2 : \mathbb{N}$, and $t > 0$, on the 2D separable monomial potential with the standard even-monomial-potentials in each coordinate, the mixed even moment satisfies

$$\langle x^{2j_1} y^{2j_2} \rangle_{2D,t} = C(k_1, k_2, j_1, j_2) \cdot t^{-j_1/k_1 - j_2/k_2}$$

where $C(k_1, k_2, j_1, j_2)$ is the composite constant

$$(2k_1)!^{j_1/k_1} \cdot (2k_2)!^{j_2/k_2} \cdot \frac{\Gamma((2j_1 + 1)/(2k_1))}{\Gamma(1/(2k_1))} \cdot \frac{\Gamma((2j_2 + 1)/(2k_2))}{\Gamma(1/(2k_2))}.$$

Plus the rescaled-Tendsto and **Asymptotics.IsEquivalent** packagings on top.

3 Proof strategy

Step 1 (factorisation): the 2D mixed-even moment factors as a product of 1D moments via the separable-observable factorisation lemma.¹ The four integrability premises specialise directly from **kth_integrable_pow_pot** (the Boltzmann weight is the $n = 0$ case).

¹**Laplace.TwoD.gibbsExpectation_separable_addSeparable.**

Steps 2-4 mirror the partition-asymptotic tide. Substitute the 1D moment closed form on each factor; peel off the t -dependence;² lift to `Tendsto` via `tendsto_const_nhds.congr'`; lift to `IsEquivalent` via the standard `IsEquivalent.refl.congr_left` pattern.

4 Roadblocks and lessons

No roadblocks. Build clean first-try, 155 lines. The mirror-shape template from the partition-asymptotic tide carried over unchanged.

Lesson: the partition $\rightarrow$ moment lift via `gibbsExpectation_separable.addSeparable` is reusable. A future numerator-asymptotic tide will compose the same way (separable factorisation + 1D numerator closed form).

5 What was Mathlib, what was new

Mathlib supplied the analytical primitives,³ and the seabed's previously formalised facts supplied everything domain-specific.⁴

What was new: the `kthKthMomentConst` definition, the four packaged theorems, and the parametric Boltzmann-integrability assembly.

6 Follow-ups

- **Generic numerator asymptotic.** The Z1 (numerator) analogue of this tide. Same template; uses the 1D `kth_moment_even` integral directly. Estimated ~ 120 lines.
- **Cross-repo promotion of the 2D-factorisation pattern.** The (partition factorisation, moment factorisation) pair on `addSeparable` potentials is now generic enough that its core could move to `lean-common`. Deferred until a second-repo consumer materialises.

²Mathlib chain: `Real.div_rpow`, `Real.rpow_add`, `Real.rpow_neg`.

³Lemmas: `Real.div_rpow`, `Real.rpow_add`, `Real.rpow_neg`, `Asymptotics.IsEquivalent.refl`, `tendsto_const_nhds.congr'`, `Filter.eventually_gt_atTop`.

⁴1D moment closed form: `gibbsExpectation_kthPotential_even`. 1D Boltzmann integrability: `kth_integrable.pow_pot`. 2D separable-observable factorisation: `gibbsExpectation_separable.addSeparable`.